PROJECT AND TEAM INFORMATION

Project Title

File Encryption & Decryption Tool

Student / Team Information

Team Number:	*T308*
Team member 1 (Team Lead)	*Rudransh Gupta –* *2510011500* *guptarudransh83@gmail.com* 
Team member 2	*Ritabrata Mandal –* *2510010593* *mandalritabrata50@gmail.com* 
Team member 3	*Nitesh Bishnoi –* *2510010972* *niteshkumarbishnoi933@gmail.com* 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Sensitive data, such as personal documents, educational records, source code, passwords, or a number of other similar items, may be contained in digital files. Such information can be easily seen by strangers if the files are stored, sent or received without protection. There is also the issue of file size – the larger the files are, the more space and bandwidth they will use, both of which could prove to be problems. The aim of this project was to develop an application, namely a File Encryption & Compression Tool, which would deal with the aforementioned problems. The application would let the user choose a file, compress the data in order to save space, and then encrypt it to prevent unauthorised access. When permission is granted, the same application can decrypt and decompress the files so that access to the data contained in them can be given. Other advantages of carrying out such a project would be gaining a greater understanding of file operations and data structures, of the use of bitwise operators, algorithms, and finally object orientated programming.

State of the Art / Current solution (1 pt)

At the moment, there are generally two different pieces of software—one for file encryption and another for file compression. Encryption enables files to be stored in a form that can only be understood when the right key or password is used, while compression allows files to be stored in a more efficient format so that the resulting file is smaller in size. As these functions are typically offered separately, it is often necessary to make use of more than one program. Certain programs that are commonly available include a large number of complicated and unnecessary features, which can confuse people who are new to the algorithms involved.

Project Goals and Milestones (2 pts)

Goals:

- *Create a functional tool for encrypting and compressing files.*
- *Set up file input and output operations.*
- *Build the compression and decompression features.*
- *Develop the encryption and decryption capabilities.*

Milestones:

- *Assess the requirements and design the overall system.*
- *Build and test the compression module.*
- *Create and test the encryption and decryption processes.*

Project Approach (3 pts)

The application is designed using C and C++. It incorporates file handling, data structures, algorithms, bit-wise operations, and object-oriented programming (OOP).

Here's how the workflow for protecting files looks:

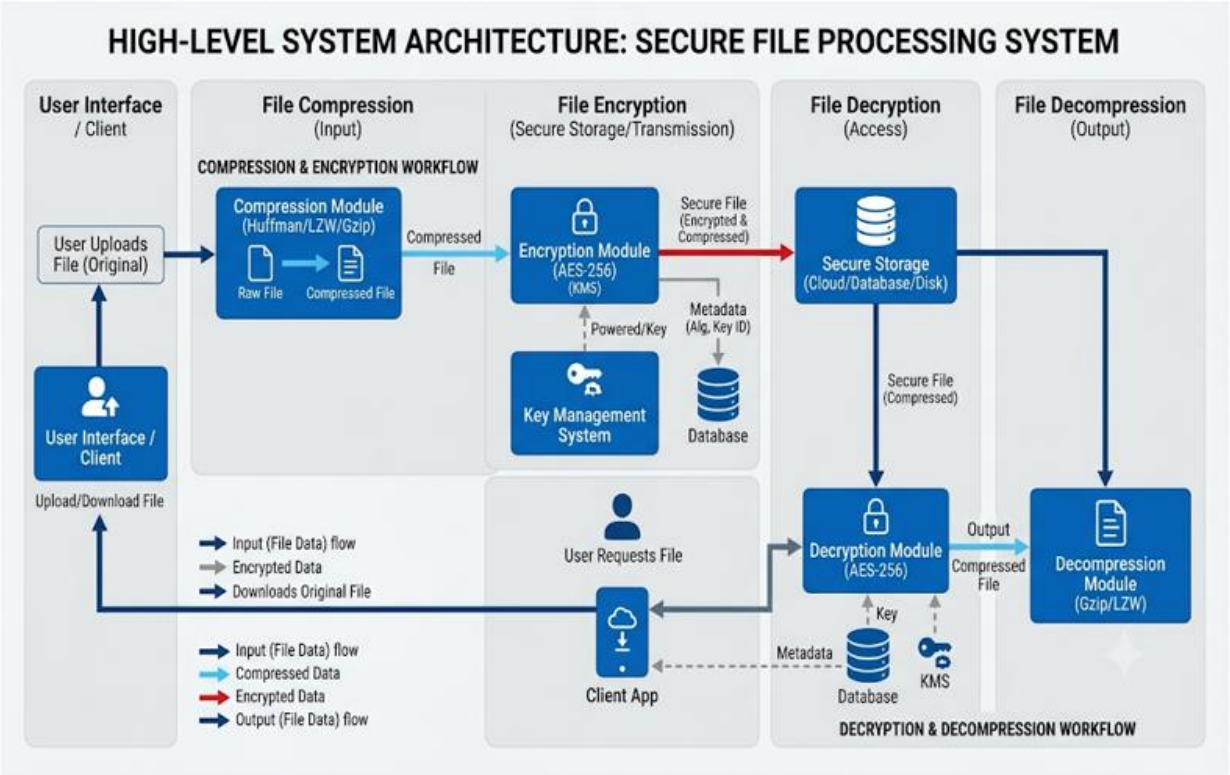
Input File -> File Handling -> Compression -> Encryption -> Protected File

For recovering the files, it goes like this:

Protected File -> File Handling -> Decryption -> Decompression -> Input File

The application includes modules for file handling, compression, and encryption/decryption, along with a user interface module to manage everything. C is utilized for basic data structure tasks and file handling, while C++ is employed for the object-oriented aspects of the application design. We've tested the application with a variety of file types, focusing on verifying the encryption and decryption processes, as well as ensuring that compression and decompression maintain file integrity and handle errors effectively.

System Architecture (High Level Diagram)



Project Outcome / Deliverables (1 pts)

The goal is to create a fully functional File Encryption & Compression Tool application that allows users to select files and perform chosen operations like encryption and compression.

Here's what we expect to deliver:

- *A working application built in C/C++.*
- *A module for handling file input and output.*
- *A module dedicated to compression.*
- *A module for encryption and decryption.*
- *A module for decompression and file recovery.*

Assumptions

The user has access to the input files they want to work with, and these files are both readable and available on the system. The goal of the program is to demonstrate the concepts of encryption and compression for educational purposes, rather than serving as a replacement for professional-grade security or compression solutions. A C/C++ compiler and development environment are included. The legal owner of the program possesses a key or password, which allows them to recover the encrypted data.